

- ESP32 (com Wi-Fi e Bluetooth integrado) - modelo DEVKITC:  
[https://pt.aliexpress.com/item/1005006651936701.html?spm=a2g0o.detail.pcDetailTopMoreOtherSeller.12.1db7ANeGANEgdi&gps-id=pcDetailTopMoreOtherSeller&scm=1007.40196.366991.0&scm\\_id=1007.40196.366991.0&scm-url=1007.40196.366991.0&pvid=86ee71ca-66aa-47d0-9b59-c49c89f24ca9&t=gps-id:pcDetailTopMoreOtherSeller,scm-url:1007.40196.366991.0,pvid:86ee71ca-66aa-47d0-9b59-c49c89f24ca9.tpp\\_buckets:668%232846%238116%232002&pdp\\_ext\\_f=%7B%22order%22%3A%22409%22%2C%22eval%22%3A%221%22%2C%22sceneId%22%3A%2230050%22%7D&pdp\\_npi=4%40dis%21BRL%2127.27%2117.27%21%21%214.50%212.85%21%402101c80217414749128488895e673c%2112000037929819276%21rec%21BR%21%21ABXZ&utparam-url=scene%3ApcDetailTopMoreOtherSeller%7Cquery\\_from%3A](https://pt.aliexpress.com/item/1005006651936701.html?spm=a2g0o.detail.pcDetailTopMoreOtherSeller.12.1db7ANeGANEgdi&gps-id=pcDetailTopMoreOtherSeller&scm=1007.40196.366991.0&scm_id=1007.40196.366991.0&scm-url=1007.40196.366991.0&pvid=86ee71ca-66aa-47d0-9b59-c49c89f24ca9&t=gps-id:pcDetailTopMoreOtherSeller,scm-url:1007.40196.366991.0,pvid:86ee71ca-66aa-47d0-9b59-c49c89f24ca9.tpp_buckets:668%232846%238116%232002&pdp_ext_f=%7B%22order%22%3A%22409%22%2C%22eval%22%3A%221%22%2C%22sceneId%22%3A%2230050%22%7D&pdp_npi=4%40dis%21BRL%2127.27%2117.27%21%21%214.50%212.85%21%402101c80217414749128488895e673c%2112000037929819276%21rec%21BR%21%21ABXZ&utparam-url=scene%3ApcDetailTopMoreOtherSeller%7Cquery_from%3A)
- Botão - CHAVE BOTÃO S/TRAVA 2P:  
[https://www.saravati.com.br/botao-push-button-250vac-1a-2t-ds-213-sem-trava-preto.html?gad\\_source=4&gad\\_campaignid=19971024214&gclid=Cj0KCQiA\\_8TJBhDNARIsAPX5qxTHTpInQf35SNmkYIAVEUrlHzbuc3dB\\_QkXY1DqNpH0DwFT1flhCusaAtxpEALw\\_wcB](https://www.saravati.com.br/botao-push-button-250vac-1a-2t-ds-213-sem-trava-preto.html?gad_source=4&gad_campaignid=19971024214&gclid=Cj0KCQiA_8TJBhDNARIsAPX5qxTHTpInQf35SNmkYIAVEUrlHzbuc3dB_QkXY1DqNpH0DwFT1flhCusaAtxpEALw_wcB)
- Módulo Leitor de Cartões RFID - RC522:  
[https://produto.mercadolivre.com.br/MLB-814816014-kit-modulo-leitor-rfid-mfrc522-mifare-JM?variation=#reco\\_item\\_pos=0&reco\\_backend=ranker\\_retrieval\\_system\\_vpp\\_v2p&reco\\_backend\\_type=low\\_level&reco\\_client=vpp-pdp-v2p-pom&reco\\_id=ffae5d6a-ce46-43c9-920d-b2649cb42a70&reco\\_model=ranker\\_entity\\_v2](https://produto.mercadolivre.com.br/MLB-814816014-kit-modulo-leitor-rfid-mfrc522-mifare-JM?variation=#reco_item_pos=0&reco_backend=ranker_retrieval_system_vpp_v2p&reco_backend_type=low_level&reco_client=vpp-pdp-v2p-pom&reco_id=ffae5d6a-ce46-43c9-920d-b2649cb42a70&reco_model=ranker_entity_v2)
- Telas - Display (LCD 16x2) I2C:  
[https://www.eletrogate.com/display-lcd-16x2-i2c-backlight-azul?utm\\_source=Site&utm\\_medium=GoogleMerchant&utm\\_campaign=GoogleMerchant&gad\\_source=4&gad\\_campaignid=22470016551&gclid=Cj0KCQiA\\_8TJBhDNARIsAPX5qxSSMLPfzVR1vywmMfOK8IKPDQY17Q-BAjfaacz5ugdOlqbqmFgWYAlj5EALw\\_wcB](https://www.eletrogate.com/display-lcd-16x2-i2c-backlight-azul?utm_source=Site&utm_medium=GoogleMerchant&utm_campaign=GoogleMerchant&gad_source=4&gad_campaignid=22470016551&gclid=Cj0KCQiA_8TJBhDNARIsAPX5qxSSMLPfzVR1vywmMfOK8IKPDQY17Q-BAjfaacz5ugdOlqbqmFgWYAlj5EALw_wcB)
- Conector placa/balança - Saída VGA - Max3232:  
[https://www.mercadolivre.com.br/conversor-rs232-ttl-max3232-serial-db9-femea-arduino-nodemcu/p/MLB32810743?pdp\\_filters=item\\_id:MLB4523421060](https://www.mercadolivre.com.br/conversor-rs232-ttl-max3232-serial-db9-femea-arduino-nodemcu/p/MLB32810743?pdp_filters=item_id:MLB4523421060)
- Resistor 10K - 10 unidades:  
[https://www.eletrogate.com/resistor-10k-1-4w-10-unidades?utm\\_source=Site&utm\\_medium=GoogleMerchant&utm\\_campaign=GoogleMerchant&gad\\_source=1&gad\\_campaignid=22470016551&gclid=Cj0KCQiA\\_8TJBhDNARIsAPX5qxT\\_ZnZCvBha1HSRHKrMpw2zoi2Xh0\\_j4dOeBI9WUVSgPIDuil6tVLEaAtPhEALw\\_wcB](https://www.eletrogate.com/resistor-10k-1-4w-10-unidades?utm_source=Site&utm_medium=GoogleMerchant&utm_campaign=GoogleMerchant&gad_source=1&gad_campaignid=22470016551&gclid=Cj0KCQiA_8TJBhDNARIsAPX5qxT_ZnZCvBha1HSRHKrMpw2zoi2Xh0_j4dOeBI9WUVSgPIDuil6tVLEaAtPhEALw_wcB)

- Resistor 4.7K - 10 unidades:  
<https://www.eletrogate.com/resistor-4k7-1-4w-10-unidades>
- Resistor 5.1K - 10 unidades:  
<https://www.eletrogate.com/resistor-5k1-1-4w-10-unidades>
- Resistor 330Ω - 10 unidades:  
<https://www.eletrogate.com/resistor-330r-1-4w-10-unidades>
- Capacitor 0.1uF:  
<https://www.eletrogate.com/capacitor-eletrolitico-1uf-x-50v>
- Capacitor Electrolítico SMD 10uF/25V:  
<https://www.eletrogate.com/capacitor-eletrolitico-10uf-x-50v>
- Módulo regulador de tensão tipo C - GT-USB-7010ASV:  
[https://pt.aliexpress.com/item/1005007205562231.html?spm=a2g0o.detail.pcDetailTopMoreOtherSeller.1.5a4dTJ8GTJ8Gz2&gps-id=pcDetailTopMoreOtherSeller&scm=1007.40050.354490.0&scm\\_id=1007.40050.354490.0&scm-url=1007.40050.354490.0&pvid=b62a1681-c8ce-4411-b673-8dc0f587e754&t=gps-id:pcDetailTopMoreOtherSeller,scm-url:1007.40050.354490.0,pvid:b62a1681-c8ce-4411-b673-8dc0f587e754.tpp\\_buckets:668%232846%238111%231996&isseo=y&pdp\\_ext\\_f=%7B%22order%3A%22%2C%22eval%3A%22%2C%22sceneId%3A%30050%22%2C%22fromPage%3A%22%2C%22recommend%3A%22%2C%22npis%3A%22%2C%22dis%21BRL%212.59%212.46%21%21%213.19%213.03%21%402101e80f17648797563131982e8ef2%2112000039798651333%21rec%21BR%21%21ABX%211%210%21n\\_tag%3A-29910%3Bd%3A8ee5d72e%3Bm03\\_new\\_user%3A-29895&utparam-url=scene%3ApcDetailTopMoreOtherSeller%7Cquery\\_from%3A%7Cx\\_object\\_id%3A1005007205562231%7C\\_p\\_origin\\_prod%3A](https://pt.aliexpress.com/item/1005007205562231.html?spm=a2g0o.detail.pcDetailTopMoreOtherSeller.1.5a4dTJ8GTJ8Gz2&gps-id=pcDetailTopMoreOtherSeller&scm=1007.40050.354490.0&scm_id=1007.40050.354490.0&scm-url=1007.40050.354490.0&pvid=b62a1681-c8ce-4411-b673-8dc0f587e754&t=gps-id:pcDetailTopMoreOtherSeller,scm-url:1007.40050.354490.0,pvid:b62a1681-c8ce-4411-b673-8dc0f587e754.tpp_buckets:668%232846%238111%231996&isseo=y&pdp_ext_f=%7B%22order%3A%22%2C%22eval%3A%22%2C%22sceneId%3A%30050%22%2C%22fromPage%3A%22%2C%22recommend%3A%22%2C%22npis%3A%22%2C%22dis%21BRL%212.59%212.46%21%21%213.19%213.03%21%402101e80f17648797563131982e8ef2%2112000039798651333%21rec%21BR%21%21ABX%211%210%21n_tag%3A-29910%3Bd%3A8ee5d72e%3Bm03_new_user%3A-29895&utparam-url=scene%3ApcDetailTopMoreOtherSeller%7Cquery_from%3A%7Cx_object_id%3A1005007205562231%7C_p_origin_prod%3A)
- Diodo tvs - modelo "TT0521SB":  
[https://www.acheicomponentes.com.br/diodo/kit-5-pecas-diodo-tt0521sb-smd-sod-882-5v-40a?parceiro=3811&srsId=AfmBOopmRrJv5\\_MlItFDYUHBjtM3FOZjTCmJr2L2QeSGrdruI\\_v-OOHdKYM](https://www.acheicomponentes.com.br/diodo/kit-5-pecas-diodo-tt0521sb-smd-sod-882-5v-40a?parceiro=3811&srsId=AfmBOopmRrJv5_MlItFDYUHBjtM3FOZjTCmJr2L2QeSGrdruI_v-OOHdKYM)
- Led verde - 3mm:  
<https://www.eletrogate.com/led-difuso-3mm-verde>
- Led vermelho - 3mm:  
<https://www.eletrogate.com/led-difuso-3mm-vermelho>